APPLIED COMPUTATIONAL THINKING (CSE 1401)

MAJOR ASSIGNMENT-1: SIMPLE ATM SIMULATOR

Institute of Technical Education & Research, SOA Deemed to be University
Branch: AIML (Artificial Intelligence and Machine Learning)

1. Objective

To implement a simple ATM simulator in Python using functions, conditional statements, return values, and loops. The program uses simple variables and does not use lists, tuples, sets, or dictionaries.

2. Approach

Task 1 - Function tasks: is_valid_amount() checks that an amount is positive, a multiple of Rs. 100, and at most Rs. 10,000. deposit() adds positive amounts and rejects others. withdraw() deducts the amount and the Rs. 10 fee only when the amount is valid and at least Rs. 500 remains.

Task 2 - Conditional tasks: check_pin() accepts PIN 4821 and otherwise returns Wrong PIN, or Card Blocked on the third wrong attempt. transaction_status() uses if-elif-else to return Invalid amount, Insufficient balance, or Approved.

Task 3 - Loop tasks: A for loop calculates the year-end balance for Rs. 10,000 invested at 6% compound interest for three years. A while loop allows up to three PIN entries. Each entry must be exactly four digits, and an entry that is not four digits still counts as an attempt. The loop uses break when the correct PIN is entered, and prints Card Blocked if all attempts fail.

Task 4 - Optional cash dispensing: note_breakdown() uses // and % to count Rs. 500 notes first and then Rs. 100 notes. It is tested with Rs. 3,700 and Rs. 800. Cash dispensing runs only after the PIN has been accepted.

3. Bank rules implemented

Rule	Value
Opening balance	Rs. 20,000
Correct PIN	4821
Maximum PIN attempts	3
Minimum balance	Rs. 500
Withdrawal fee	Rs. 10 per withdrawal
Valid withdrawal amount	Positive multiple of Rs. 100; maximum Rs. 10,000
Available notes	Rs. 500 and Rs. 100

4. Test cases and observed results

Test	Expected result	Observed result
is_valid_amount(1000)	True	True
is_valid_amount(1050)	False	False
deposit(20000, 1500)	21500	21500
deposit(20000, -500)	20000	20000
withdraw(20000, 2000)	17990	17990
withdraw(20000, 1050)	20000 (invalid)	20000
check_pin("4821", 1)	Accepted	Accepted
check_pin("1111", 3)	Card Blocked	Card Blocked
transaction_status(20000, 1000)	Approved	Approved
transaction_status(20000, 1050)	Invalid amount	Invalid amount
transaction_status(600, 1000)	Insufficient balance	Insufficient balance
note_breakdown(3700)	7 x Rs. 500, 2 x Rs. 100	7 and 2 notes
note_breakdown(800)	1 x Rs. 500, 3 x Rs. 100	1 and 3 notes
PIN entries: 4855, 5584, 6654	Card Blocked after 3 attempts	Card Blocked after 3 attempts

5. Compound-interest result

Using annual compounding at 6%, the closing balances are Rs. 10,600.00 after Year 1, Rs. 11,236.00 after Year 2, and Rs. 11,910.16 after Year 3.

6. Conclusion

The program implements the required functions, conditional checks, compound-interest loop, PIN-entry loop, and optional note breakdown. The test cases cover valid and invalid inputs, successful and failed PIN attempts, and insufficient balance. Cash dispensing is skipped when the card is blocked.